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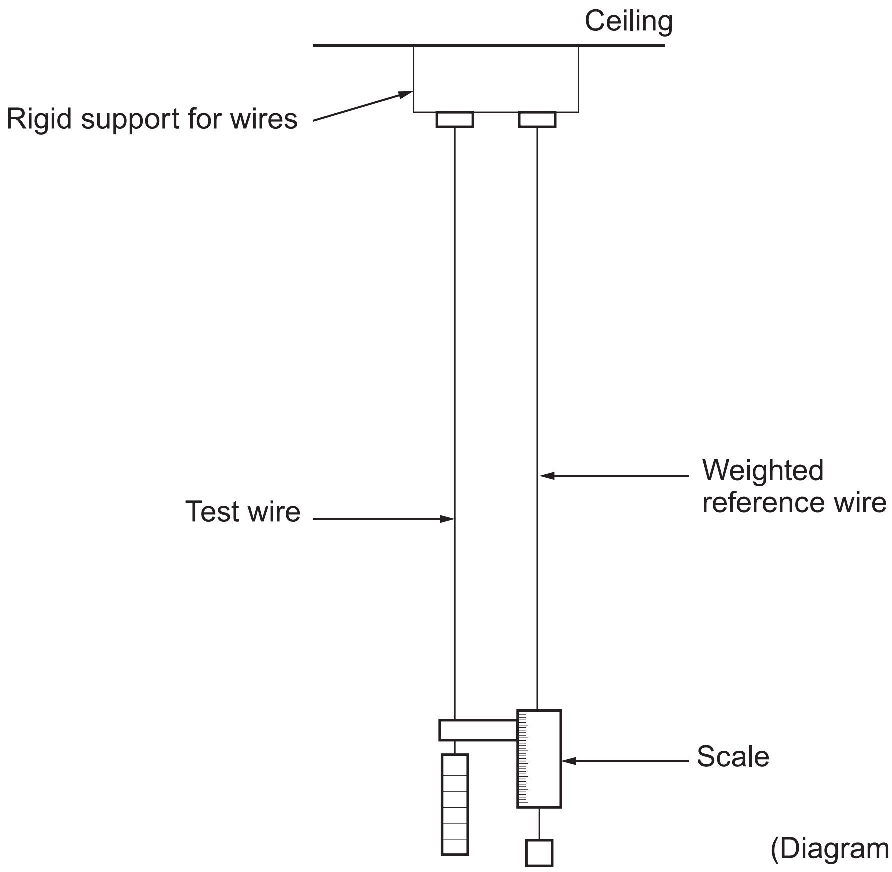
4. The following extract from a Physics text book describes a method for determining the Young modulus of a metal in the form of a wire.

Apparatus

A long test wire and a reference wire of the same length and material are suspended from a common rigid support. This minimises the effect of temperature and the movement of the support. The scale for measuring extensions is on the reference wire and a weight is placed on it to keep it taut and kink free. This way if the test wire pulls the support downwards, the reference wire and scale move with it. The scale will therefore read only the extension of the test wire.



(Diagram not to scale)

Method

Measure the extension for increasing loads ensuring that the wire remains within its *elastic limit.* Take repeat readings during the removal of the load to provide a mean value for the extension. Measure the diameter of the test wire at several different places using a micrometer. Measure the length of the test wire with a ruler.

(a) (i) Explain how the effect of a change in temperature on the wire is minimised. [2]

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(ii) State what is meant by the term *elastic limit* **and** explain how an experimenter would know whether or not the test wire has extended beyond its elastic limit. [2]

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(iii) The text book also states:

'The apparatus and procedures are designed carefully to minimise uncertainties when taking measurements.'

I. State why a long wire is used rather than a short one. [1]

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II. Why is the diameter of the wire measured at several different places? [1]

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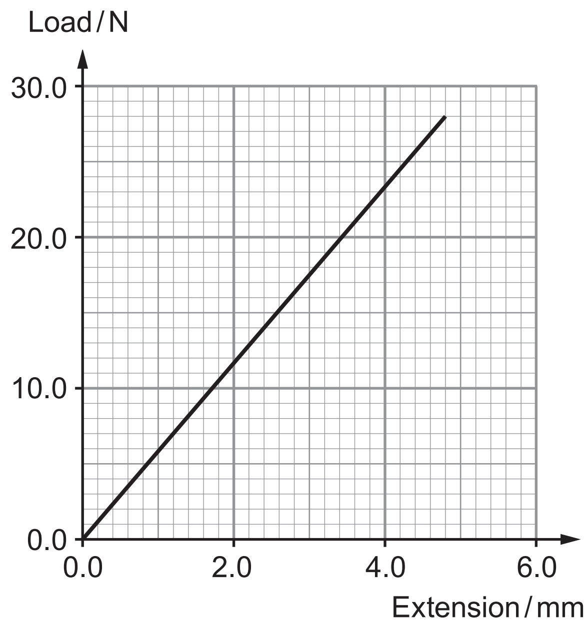
(b) A typical graph showing the results of such an experiment is given below. The original length of the wire is 2.40 m and the mean value of its **diameter** is 0.32 mm.



(i) Use the graph and the measurements given to determine the Young modulus of the material of the wire. Give your answer to an appropriate number of significant figures. [4]

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(ii) Calculate the energy stored in the wire when it is extended by 2.4 mm. [2]

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